IoMT Analysis Tool Requirements

UW-Madison Capstone Fall 2026

This document details the user requirements for the IoMT Analysis Tool that will be created as part of the UW-Madison Fall 2026 Capstone course. All requirements should be considered and designed for throughout the duration of this project. Requirements are subject to change as the project develops.

Please consider the diagrams below detailing how the IoMT Analysis Tool fits into the overall patient monitoring system.

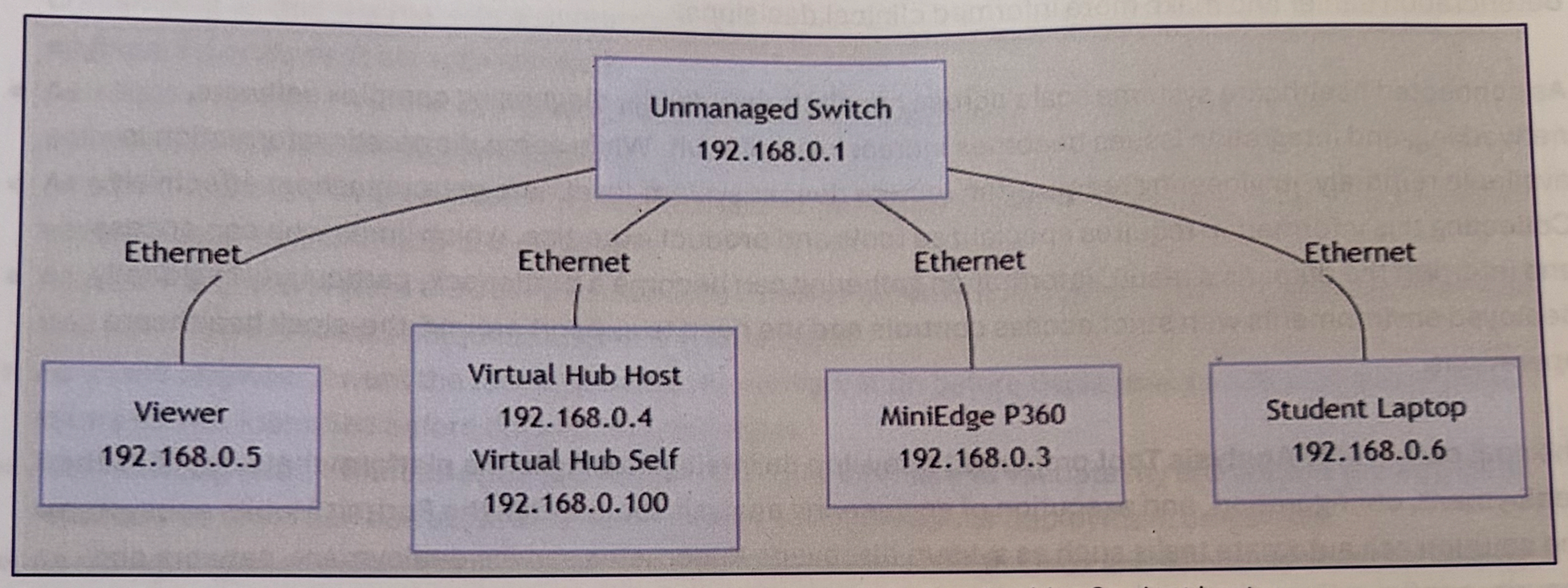


Figure 1: Portrait Mobile System setup. Static IPs already set on all devices besides Student Laptop

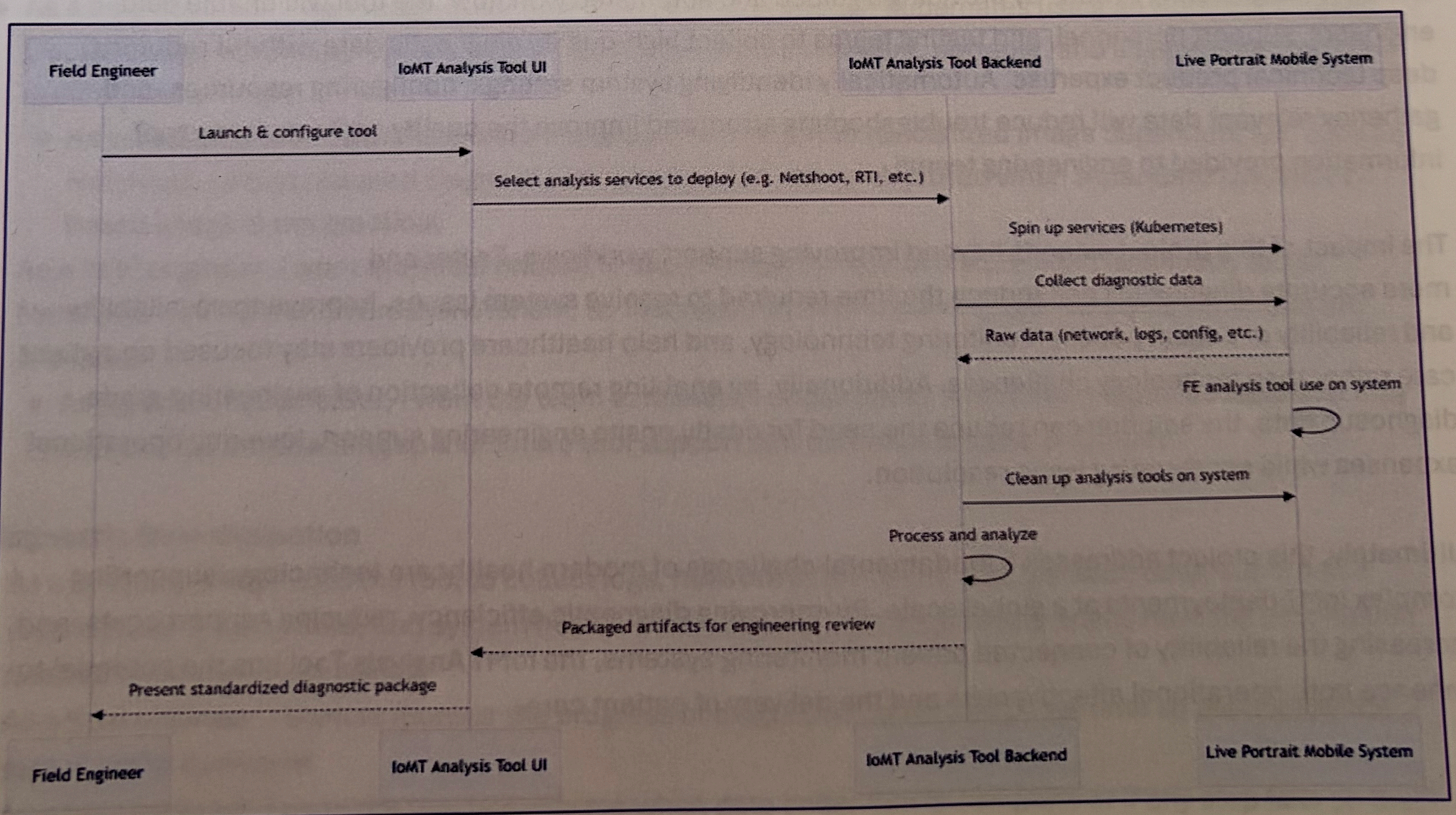


Figure 2: IoMT Analysis Tool high-level workflow.

Project Description

The **Internet of Medical Things (IoMT)** is a connected ecosystem of medical devices, software applications, cloud services, and healthcare systems that collect, share, and analyze clinical data in real time. As healthcare becomes increasingly connected, ensuring the reliability, security, and performance of these systems is essential to clinical operations and patient care.

GE HealthCare's **Portrait Mobile** platform is a wearable continuous patient monitoring solution that enables clinicians to monitor vital signs remotely while supporting patient mobility throughout the hospital. By continuously capturing and communicating patient data, Portrait Mobile helps care teams detect patient deterioration earlier and make more informed clinical decisions.

As connected healthcare systems scale across hospitals worldwide, diagnosing complex software, networking, and integration issues becomes increasingly difficult. While some diagnostic information is available remotely, engineering teams often require deeper system-level data to troubleshoot effectively. Collecting this information requires specialized tools and product expertise, which limits who can access and interpret the data. As a result, information gathering can become a bottleneck, particularly in globally deployed environments with strict access controls and the need to support around-the-clock healthcare operations.

The goal of the **IoMT Analysis Tool** project is to develop an intelligent diagnostic platform that simplifies the deployment, configuration, and execution of engineering analysis tools within the Portrait Mobile ecosystem. The solution will automate tasks such as system discovery, Kubernetes service deployment, network and security configuration, diagnostic data collection, and packaging analysis artifacts into a standardized format for engineering review. By providing a guided and automated workflow, the tool will enable field engineers, support personnel, and testing teams to collect high-quality diagnostic data without requiring deep technical product expertise. Automatically identifying system settings, configuring resources, and gathering relevant data will reduce troubleshooting errors and improve the quality and consistency of information provided to engineering teams.

The impact of this project extends beyond improving support workflows. Faster and more accurate diagnostics can reduce the time required to resolve system issues, improve the availability and reliability of critical patient monitoring technology, and help healthcare providers stay focused on patient care rather than technology challenges. Additionally, by enabling remote collection of engineering-grade diagnostic data, the solution can reduce the need for costly onsite engineering support, lowering operational expenses while accelerating issue resolution.

Ultimately, this project addresses a fundamental challenge of modern healthcare technology: supporting complex IoMT deployments at a global scale. By improving diagnostic efficiency, reducing support costs, and increasing the reliability of connected patient monitoring systems, the IoMT Analysis Tool has the potential to enhance both operational effectiveness and the delivery of patient care.

Requirements

Deployment & Configuration

- As a field engineer, I want to deploy engineering analysis tools to a live Portrait Mobile environment so that diagnostic data can be collected without manually installing software on each system.
- As a field engineer, I want to be able to use the deployed engineering analysis tools on the live Portrait Mobile system.
- As a field engineer, I want the system to automatically discover available Portrait Mobile components (viewer, hub, backend services, Kubernetes nodes) so that I do not need to manually identify each device.
- As a field engineer, I want to provide the platform's connection information (such as an IP address and credentials) so that the tool can automatically identify network and system settings and configure engineering analysis tools appropriately.
- As a field engineer, I want to select which analysis tools or diagnostics to run so that I can tailor data collection to the issue being investigated.
- As a field engineer, I want a graphical user interface to configure analysis tools so that I can use the system without requiring command-line expertise.
- As a field engineer, I want the ability to manually override automatically generated configuration values so that I can accommodate unique customer environments.
- As a field engineer, I want the tool to validate my configuration before deployment so that common setup errors can be identified before data collection begins.
- As a field engineer, I want all open text fields in the user interface to validate my entries and provide clear feedback so that I can correct invalid inputs before configuration or deployment continues.
- As a field engineer, I want the graphical user interface to run on both Linux and Windows workstations so that I can use the tool in supported field and engineering environments.
- As a field engineer, I want diagnostic tools to be deployed through Kubernetes using an Ubuntu 22-based Docker image where appropriate so that deployments are consistent, portable, and aligned with the target system environment.
    - As a field engineer, I want approved exceptions for tools with specialized image dependencies, such as Netshoot, so that required diagnostic capabilities can still be deployed when a standard Ubuntu 22-based image is not practical.
- As a field engineer, I want the initial release to support deployment of the provided diagnostic tools: Netshoot, UMF RTI, and Wireshark/tshark, so that required network and system analysis workflows are available.
    - As a project stakeholder, I want the team to research and propose additional diagnostic tools that may improve troubleshooting so that future tool support can be evaluated and prioritized.

Diagnostic Data Collection

- As a field engineer, I want the tool to collect logs, network information, configuration data, Kubernetes diagnostics, Grafana data, and system health information so that engineering teams have the information needed to troubleshoot issues.
- As a field engineer, I want to monitor the progress of diagnostic collection in real time so that I know the status of the operation.
- As a field engineer, I want the tool to notify me when data collection is complete or if any step fails so that I can take corrective action.

- As a field engineer, I want deployed diagnostic services to be automatically removed after data collection & use is complete so that the customer environment is returned to its original state.
- As a field engineer, I want the tool to perform cleanup when I end a session and prompt me to choose the appropriate cleanup action after an unexpected crash or interruption so that temporary deployments, files, and system changes are removed or preserved intentionally and the customer environment is left in a known clean state.

Analysis & Reporting

- As a field engineer, I want collected diagnostic data to be packaged into a single transferable file so that I can easily provide it to engineering teams.
- As a field engineer, I want analysis artifacts to follow a standardized format so that engineering teams receive consistent diagnostic packages regardless of who collected them.
- As a field engineer, I want diagnostic packages to include metadata about the environment being analyzed so that engineering teams have sufficient context for troubleshooting.
- As a field engineer, I want the tool to generate a high-level summary of detected issues, warnings, and system conditions so that I can quickly understand the results (stretch).

Security

- As a field engineer, I want authentication to be required before accessing the tool so that diagnostic functionality is restricted to authorized users.
- As a field engineer, I want my UI login session to automatically lock or expire after 5 minutes of inactivity while backend diagnostic processes continue running so that unauthorized users cannot access diagnostic information from an unattended workstation without interrupting active data collection.
- As a field engineer, I want the tool to identify where PHI or PII may be collected within the diagnostic workflow so that potential privacy-sensitive data collection points are documented.
- As a field engineer, I want all tool actions and configuration changes to be logged so that usage can be audited if necessary.

Usability

- As a field engineer, I want the user interface to follow GE HealthCare branding guidelines so that it provides a familiar and professional experience.
- As a field engineer, I want the user interface to remain readable from approximately 1.5 feet away so that I can comfortably operate the system in field environments.
- As a field engineer, I want guided workflows and contextual help so that I can successfully collect diagnostic data without requiring deep Portrait Mobile expertise.
- As a field engineer, I want clear error messages and recommended corrective actions so that I can resolve issues without extensive troubleshooting knowledge.

Reliability

- As a field engineer, I want the tool to continue operating when disconnected from the internet so that it can be used in restricted hospital networks.
- As a field engineer, I want partial results to be preserved if a collection process fails so that previously gathered information is not lost.

- As a field engineer, I want the tool to complete the diagnostic workflow with minimal impact to the live Portrait Mobile system so that patient monitoring operations are not disrupted.
- As a field engineer, I want the tool to enforce configurable CPU limits, memory limits, throttling controls, and file size limits during diagnostic collection so that the tool does not consume excessive system resources or generate unmanageable output files in the live customer environment.